Bali Dynasty Resort, Kuta Bali Hotel, Indonesia

SMARTGENCON

International Conference on Smart Generation Computing, Communication and Networking

4th SMARTGENCON 2025

10th – 11th October 2025

Jointly Hosted By

UNIVERSITAS UDAYANA

K. L. E. Society's K.L.E. Institute of Technology, Hubballi-580027

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2025 International Conference on Smart Generation Communication, Computing and Networking

10 – 11 October, 2025

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Message

Dr. Basavaraj Anami
Registrar
KLE Technological University
Hubballi

With immense pleasure, I am writing this message that KLE Institute of Technology, Gokul, Hubballi-580027 in association with Udayana University, Bali has jointly organized 4th SMARTGENCON 2025 during 10th -11th October 2025.

SMARTGENCON has rapidly emerged as a global platform for exchanging cutting-edge ideas and breakthroughs in smart technologies. This year's edition continues that tradition, bringing together an inspiring mix of researchers, industry leaders and innovators from across the world to share their latest work and to foster collaborations.

I warmly acknowledge the efforts of our organizing committee, advisory board, reviewers and volunteers who have worked tirelessly to ensure a high-quality, enriching experience for all participants.

On behalf of the entire team, I extend my best wishes for a highly successful conference. May SMARTGENCON 2025 spark new ideas, partnerships and pathways for advancing smart generation computing, communication and networking worldwide.

Message

Dr. Manu T M
Principal,
KLE Institute of Technology,
Hubballi 580027, Karnataka, India

It is with great pride and joy that I welcome you all to the 4th International Conference on Smart Generation Computing, Communication and Networking (SMARTGENCON 2025) being held from 10th–12th October 2025 at Bali, Indonesia.

As the General Chair of this conference and the principal of the host institution KLE Institute of Technology Hubballi Karnataka India, it is truly gratifying to see SMARTGENCON evolve into a premier platform where academicians, researchers, industry professionals, and students from across the globe convene to share knowledge, showcase innovations, and discuss the future of smart technologies.

This conference reflects our institution's commitment to fostering excellence in research and innovation. I am confident that the deliberations and exchanges taking place over these three days will ignite new collaborations, inspire pioneering ideas, and contribute significantly to the advancement of computing, communication, and networking.

On behalf of the organizing committee and our host college, I extend my heartfelt thanks to all keynote speakers, authors, delegates, sponsors, and volunteers for their invaluable contributions.

I wish SMARTGENCON 2025 grand success and hope that every participant leaves with enriching experiences and meaningful connections.

Message

Dr. Rajesh Yakkundimath
Convenor
KLE Institute of Technology
Hubballi 580027, Karnataka, India

It gives me immense pleasure to welcome you all to the 4th International Conference on Smart Generation Computing, Communication and Networking (SMARTGENCON 2025), hosted by KLE Institute of Technology, Hubballi, in collaboration with our esteemed partners, from 10th–12th October 2025 at Bali, Indonesia.

As Head of the Department of Computer Science & Engineering and a proud member of the conference team, I am honored that our institution is playing a key role in bringing together an outstanding gathering of researchers, academicians, industry experts and students from around the globe.

SMARTGENCON 2025 reflects our department's commitment to academic excellence, innovation and global collaboration. The technical sessions, keynote addresses and panel discussions will provide a fertile ground for exchanging ideas, showcasing research outcomes and building new networks.

On behalf of the Department of CSE and the host institution, I extend my heartfelt gratitude to all the authors, reviewers, keynote speakers, sponsors and volunteers who have contributed to making this event a reality.

I wish SMARTGENCON 2025 great success and hope every participant finds it intellectually stimulating and professionally rewarding.

Message

Dr. Shridhar Allagi
Publication Chair
KLE Institute of Technology,
Hubballi, Karnataka, India

It is my privilege and pleasure to welcome all participants to the 4th International Conference on Smart Generation Computing, Communication and Networking (SMARTGENCON 2025) being held from 10th–12th October 2025 at Bali, Indonesia.

As Publication Chair, I am delighted to see this conference grow into a vibrant platform that connects researchers, academicians, professionals and students from around the world to exchange ideas, present their latest work and explore new frontiers in smart technologies.

This year's edition of SMARTGENCON reflects the dedication of our organising committee and the strong support of our host institution, partners and volunteers. We have curated an exciting programme of keynote addresses, technical sessions and panel discussions that promise to stimulate thought and collaboration.

On behalf of the organising team, I thank our keynote speakers, reviewers, authors, sponsors and delegates for their invaluable contributions. I am confident that the discussions here will lead to meaningful partnerships and innovations that advance the fields of computing, communication and networking.

I extend my best wishes for a successful and enriching SMARTGENCON 2025 to every participant.

Keynote Speaker

Dr. Nyoman Gunantara

Nyoman Gunantara, joined with Universitas Udayana (Unud), Bali as lecturer since 2001. He holds bachelor's degree in electrical engineering from Universitas Brawijaya (UB), Master and Doctor degrees in electrical engineering from Institut Teknologi Sepuluh Nopember (ITS). He has 20 years of teaching and research experience. His research interests include wireless communications, ad-hoc network, quality network, and optimization. He is a member of the IEEE

Keynote Speaker

Mr. Sahktiprasad Anami,
Manager-SOC | HEAD- Cybersecurity Operations @ GRAMAX-GMR Group.

Mr. Shaktiprasad Anami is a seasoned cybersecurity professional with rich experience in managing global Security Operations Centers (SOC) and leading cyber defense strategies across diverse sectors including banking, defense, manufacturing, pharmaceutical, and education. He currently serves as Manager-SOC and Head of Cybersecurity Operations at GRAMAX – GMR Group.

With proven expertise in SIEM & SOAR (IBM Qradar, Splunk, Solarwinds, WAZUH), EDR/XDR solutions, threat intelligence, ransomware mitigation, cloud security, and incident response, he has consistently strengthened organizations' security posture and resilience. He is also proficient in phishing analysis, malware reverse engineering, data loss prevention, and application/server security.

An engineering graduate from Visvesvaraya Technological University, Mr. Anami holds multiple certifications including CCNA, CEH, Microsoft AZ-500, Fortinet NSE 1 & 2, Digital Forensics, and is recognized as a Certified Cyber Hygiene Expert by the Government of India.

Known for his hands-on approach, leadership in SOC operations, and ability to design robust cybersecurity frameworks, he continues to drive cyber resilience and digital trust in critical infrastructures.

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SMARTGENCON 2025 Programme Schedule

2025 International Conference on Smart Generation Communication, Computing and Networking Venue: Discovery Kartika Plaza Hotel, Kuta, Bali 80361, Indonesia 10th and 11th OCTOBER 2025 Conference Main Schedule			
	Time	**Events**	**Place**
Day 1: **10 OCT 2025**	**8.45 am -10.45am**	Session -I	
	10.45am - 10.50am	Welcome By Dr. Rajesh Yakkundimath- HoD CSE, KLEIT Hubballi	**RAMA Hall**
	10.50am – 10.55am	Welcome and Release of Conference Proceedings	
	10.55 am -11.10 am	Brief about Conference by General Chair – **Dr. Manu T M , Principal, KLE Institute of Technology Hubballi**	
	11.20 am -11.35 am	Address by Chief Guest – Dr. Basavaraj Anami, Registrar, KLE Technological University, Hubballi	
	11.35 am -11.45 am	Address by Guest of Honor- Dr. Chanakya Kumar Jha Director Planning Siddhant Group of Institutions, Pune India	
	11.45 am -12.15 pm	Vote of thanks by – Dr. Shridhar Allagi, Co-Convenor , KLEIT Hubballi	
	12.15 pm -12.45 pm	Keynote By : **Ir Komang Oka Saptutra (Udyana University, Bali)**	
	12.45 pm- 1.30 pm	Keynote By : **Mr. Sahktiprasad Anami**, Manager-SOC | HEAD- Cybersecurity Operations @ GRAMAX-GMR Group.	
	1.30pm to 3.00pm	Lunch	
	4.00 pm - 6.00 pm	Session –II	
	4.00 pm - 6.00 pm	Session –III	

SMARTGENCON 2025 Programme Schedule

Day 2: 11 OCT 2025	**8.45 am - 10.45am**	Session –I	**Online**
	10.45 am -11.00 am	Tea break	
	11.00am - 12.00 pm	Keynote By – Mr. Anil Kumar Jonnalagadda (Independent AI/ML Researcher, USA)	
	1.05 pm – 1.45 pm	Lunch	
	1.45 pm - 3.45 pm	Session –II	
	3.45 pm – 4.00 pm	Tea Break	
	4.00 pm - 6.00 pm	Session -III	

Abstracts

Paper ID: 1033

Stree: A Smart Solution for Women's safety in emergency response

Shilpa R (Vidyavardhaka College of Engineering)*

Abstract: The paper presents an innovative security device which swiftly aids women who find themselves in emergency situations. A comprehensive safety solution emerges through the combination of a Raspberry Pi 3B+ with GSM module and GPS module together with buzzer and push button. The device activates its tracking functionality when a user holds the button for an extended time and collects location data and camera images that get sent to previously registered SMS and email recipients. The paper explores the system structure and describes how it incorporates different sensors as well as modern technological components. The article presents operational insights regarding how the device works while demonstrating potential enhancements it could receive in the future. The primary objective of this development is to provide women with a dependable device which simultaneously maintains their safety while delivering instant assistance during emergencies.

Paper ID: 1034

Embedded Linux on Energy-Efficient Boards: Transition from MAGIK2 to Next-Gen Platforms

Shilpa R (Vidyavardhaka College of Engineering)*

Abstract: The Android operating system, which is modeled after Linux, has garnered quite a following for its open-source nature. The fact that it's open source allows users to customize their interface according to their preference; this flexibility how- ever also renders it easily adaptable. ARM is a family of reduced instruction computer architecture for computer processors that can be configured for various environments. The main idea of this project is to port Android 7.1.2 (Nougat) over Linux kernel 4.9.17 onto the ARM-based i.MX6 processor. The MAGIK2 (Media Accelerated Graphical Innovation kit) board is used for porting Android OS which is based on i.MX6 dual-core processor. Porting is carried out by customizing the source code, compiling the code, flashing the binary images, and finally booting the system. Android is ported onto the MAGIK2 board to drive an LVDS display device in the present work.

Paper ID: 1119

Retail Theft Detection Using Machine Learning for Enhanced Security Surveillance

Livini Wijerathne (University of Westminster)*; Chamupathi Hettige (Informatics Insititute of Technology)

Abstract: Shoplifting remains a persistent issue in the retail industry, contributing to significant financial losses and security challenges. Conventional surveillance systems rely on human monitoring and are often inefficient due to fatigue, subjectivity, and limitations in real-time analysis. To address these shortcomings, the proposed system introduces an advanced deep learning-based shoplifting detection framework capable of analyzing surveillance footage with high accuracy. The system employs a variant of Convolutional Neural Networks (CNNs) based on MobileNetV2 for efficient feature extraction and Bidirectional Long Short-Term Memory (BiLSTM) networks to capture sequential patterns in human behavior, enabling precise identification of suspicious activities. Through extensive evaluation on a curated dataset, the proposed model achieved an accuracy of 93%, demonstrating its effectiveness in distinguishing normal shopping behaviors from shoplifting incidents. By automating theft detection and reducing reliance on manual surveillance, the proposed approach enhances security measures in retail environments, offering a scalable and efficient solution for loss prevention.

Paper ID: 1120

Career Path Recommendation System Using Ensemble Learning

Oshini Fonseka (University of Westminster)*; Gigara Hettige (Informatics Institute of Technology)

***Abstract**:* Choosing an appropriate career plan is a challenging task for students due to the vast number of available career options and a lack of proper guidance tailored to individual skills and interests. Traditional career counseling methods rely on subjective assessments and expert opinions, often failing to provide personalized and data-driven recommendations. This research proposes a machine learning based career path recommendation system that predicts suitable career choices by analyzing student performance. The system employs a Stacking Classifier with base models including LightGBM, Random Forest, XGBoost, and Support Vector Classifier (SVC), optimized using Bayesian hyperparameter tuning and manual refinement. Data preprocessing, feature engineering, and model optimization techniques were implemented to enhance predictive accuracy. Model evaluation using a train-test split strategy demonstrates an overall accuracy of 88%, with precision and recall scores indicating strong predictive performance.

Paper ID: 1125

Intelligent Cricket Highlights: A Multi-Modal Approach Using Video and Audio Data

Prajwal L (PES University)*; Sujay Gudur (PES University); M. Surya Gagan Reddy (PES University); Anushree M Sulibhavi (PES University); Dinesh Singh (PES University)

***Abstract**:* In the world of cricket, the demand for quickaccess to key moments from lengthy matches has risensignificantly. Traditional methods of highlight generationare often labor-intensive, requiring manual effort to siftthrough hours of footage.This paper proposes anautomated system for generating cricket highlights byleveraging video and audio analysis. The systemintegrates Optical Character Recognition (OCR) forscoreboard detection, audio processing to identifysignificant events, and a Genetic Algorithm (GA) foroptimizing event sequencing based on user preferences.The proposed approach ensures that highlights aretailored, offering personalized viewing experiences.Experimental results demonstrate the effectiveness of thesystem, with high accuracy in event detection andhighlight quality, along with positive user feedback. Thesystem's scalability and adaptability to different cricketformats and conditions highlight its potential forwidespread use in sports media.Future improvements willfocus on reducing the system's reliance on manual inputand enhancing event detection using deep learningtechniques

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Paper ID: 1133

Image-Inspired Melodies: An Image Based Music Recommendation System

Neelalohith R Kashyap (PES University)*; Nishanth R Kashyap (PES University); K Virupakshi (PES University); Gajendra S (PES University); Dinesh Singh (PES University)

***Abstract**:* Gradual Change of trend in recommendation systems become necessarily important for customer satisfaction on music streaming platforms. This paper introduces us to a novel music recommendation system that utilizes the facial emotion of the user as well as emotion of each song in order to recommend songs as well as generating a personalized user music playlist which is in perfect synchronization with the user's mood. Utilization of such novel recommendation systems is necessary in order to increase customer satisfaction as well as ensure that the customers are active on music streaming platforms which can globally benefit these music streaming platforms from a business standpoint as well as from a technical standpoint.

Paper ID: 1213

Enhancing Enterprise Infrastructure Security Through Machine Learning Technique

Naved Khan (Amity Institute of technology)*

***Abstract**:* The sophistication of current internet attacks has led firms to adopt ML and AI in addition to traditional security measures. By analyzing huge datasets, ML algorithms can detect anomalies that deviate from normal behavior patterns, while AI-powered systems integrate such threat intelligence information and ana lyze it so as to recognize the emerging threats beforehand. By using machine learning (ML) technology, behavioral analysis can identify unusual activities and inside jobs; predictive analytics help forecast potential risks based on historical data. These responses are automated, without any human intervention to miti gate incidents rapidly and adaptive security controls change with ever-changing threat environments. In finance or e-commerce sector, ML together with AI also assists in fraud detection. Nevertheless, successful integration requires strong governance, risk management and compliance practices as well as continued mon itoring and enhancement of ML and AI models. In summary, the inclusion of ML and AI in enterprise security architecture helps organizations better protect themselves against evolving cyber threats.

Paper ID: 1229

AI Powered Centralized Travel Agency Platform for Sri Lanka

Ravihara Gunathilaka (Sri Lanka Institute of Information Technology)*; Hasini Hewage (Sri Lanka Institute of Information Technology); Kalana Chamod (Sri Lanka Institute of Information Technology); Devin Sandanayake (Sri Lanka Institute of Information Technology); Uthpala Samarakoon (Sri Lanka Institute of Information Technology); Wishalya Tissera (Sri Lanka Institute of Information Technology)

***Abstract**:* This Tourism plays a crucial role in driving Sri Lanka's economic growth, contributing significantly to job creation and national revenue. The current system for travel planning in Sri Lanka has significant shortcomings that reduce its efficiency and effectiveness. One primary issue is that travelers must visit multiple websites and agencies to collect proper travel information. Those websites provide statics and pre-planned details which are not completely suitable for the tourists' needs. Additionally, navigating to a new country usually requires the constant support of a tour guide to ensure an enjoyable experience because journeying alone and navigating in an unfamiliar country can be challenging and filled with risks. To address the challenges faced by Sri Lanka's tourism industry, a new AI-based centralized system for travel agencies in Sri Lanka is being suggested. This system will be a complete platform for tourists, offering personalized travel suggestions and strong planning recommendations that match each individual need by analyzing them deeply. Additionally, the platform will send automatic safety real-time mobile alerts and warnings by providing awareness of the location by tracking the tourists' current locations. Tourists can use AR to obtain detailed information about specific places that they hope to visit, providing a richer and more informative experience without the need for guides. The AI-based centralized travel platform integrates using Spring Boot, Firebase, and Next.js, the platform leverages Google Maps, OpenWeatherMap, and Google Cloud Vision for enhanced navigation and personalized recommendations. Accuracy is increased by integrating data from variety of sources, including tourist and travel agency surveys, Sri Lanka tourism websites, and national databases. Future development will focus on refining AI models and incorporating user feedback to ensure continuous improvement and adaptability.

Paper ID: 1237

AI-Powered Tea Factory Management System

Kavindu Gunawardane (Sri lanka institute of information technology)*; Sathsara Wijekulasuriya (Sri lanka institute of information technology); Ayesha Jinadasa (Sri lanka institute of information technology); Thevin Abeykoon (Sri lanka institute of information technology); Uthpala Samarakoon (Sri lanka institute of information technology); Wishalya Tissera (Sri lanka institute of information technology)

***Abstract**:* The Sri Lankan tea industry is a cornerstone of the country's economy, contributing significantly to export revenues and employment. However, traditional tea production methods are fraught with inefficiencies, particularly in yield prediction, demand forecasting, quality control, and disease identification. These inefficiencies stem from reliance on outdated manual processes, subjective human judgment, and the lack of advanced data-driven techniques. This research proposes an AI-powered tea factory management system designed to address these challenges by incorporating machine learning models for yield prediction, AI-based demand forecasting, computer vision for automated tea quality assessment, and deep learning for real-time tea plant disease identification.The system processes data from diverse sources, including historical production records, meteorological data, market trends, and real-time plant health diagnostics, to optimize decision-making and operational efficiency. Additionally, mobile and web applications facilitate real-time tea grading, demand forecasting, and disease detection, enabling tea manufacturers to ensure product consistency while proactively managing production risks.Experimental results demonstrate that AI-powered systems significantly enhance accuracy in yield forecasting, demand estimation, quality grading, and disease detection, leading to improved efficiency and reduced losses. The adoption of AIdriven automation fosters sustainability and increases competitiveness in global markets. Future research directions include the integration of IoT-based monitoring systems and AIdriven precision agriculture techniques to further enhance productivity in the tea industry.

Paper ID: 1259

The Cost-Effective Mechanism for Enhancing Solar Panel Performance By Temperature Controlling Using Radiator and Airflow

Poonam Chaudhari (Yeshwantrao Chavan College of Engg)*

***Abstract**:* The Renewable Energy is on its Zenith point to flourish and in accordance with a revolutionary change in Technology has been made. Solar Energy is one of the types of Clean Energy which is widely used throughout Globe by Installation of SPV System. However, this system has to face so many challenges, one being the effect of Temperature on performance of the System. This research explores an innovative approach to cooling solar panels using a radiator-based water-cooling system supplemented by natural airflow. The system leverages the principles of heat exchange to reduce the operating temperature of photovoltaic (PV) cells, thereby improving efficiency and extending lifespan.

Paper ID: 1282

Learning System To Enhance Reading Skills In Children With Dyslexia

Kaveesha Wijethunga (SLIIT)*; Sithum Bandara (SLIIT); Kavindu Thennakoon (SLIIT); Hansaja Gunasekara (SLIIT); Samantha Rajapaksha (SLIIT); Nelum Amarasena (SLIIT)

Abstract*:* This research aims to enhance the learning experience for dyslexic children that affects ability to read, write, pronounce words, and maintain focus. According to the research the mobile application designed to improve their writing, reading, pronunciation, and cognitive focus. The writing module utilizes touch-responsive technology to assist dyslexic children in writing words and numbers, providing real time feedback, error detection, and corrective recommendations. To address reading challenges, the application incorporates readability tools that modify text elements such as fonts, spacing, and color contrast, making reading more accessible. Real-time text analysis and customization ensure that children receive personalized support tailored to their needs. The research also focuses on advancing cognitive development by integrating engaging graphics, aural cues, and interactive content that reduce visual discomfort and improve attention. The system supports multiple languages, making it accessible to a diverse group of dyslexic learners. In addition, the pronunciation and literacy module (ReadBuddy) help children develop proper pronunciation skills through digital imagery and word segmentation. By providing structured feedback and training, the system adapts to each child's learning pace, fostering gradual literacy improvement. To ensure the application's effectiveness, the research includes controlled trials and user feedback analysis, assessing improvements in writing, reading fluency, comprehension, and cognitive abilities. Findings from these evaluations will guide further optimizations, ensuring the system remains a valuable and adaptable learning tool for dyslexic children. By integrating innovative technology and evidence-based strategies, this research aims to create a comprehensive, interactive, and accessible learning environment that supports dyslexic children in overcoming their educational challenges and achieving academic success.

Paper ID: 1350

Automating Third Umpire Decision Making Process Using Deep Learning Techniques

Janith Harshana (SLIIT)*; Navindya Jayathilaka (SLIIT); Nethmini Senarathna (SLIIT); Pehan Dassanayake (SLIIT); Wishalya Tissera (SLIIT); Chathushki Chathumali (SLIIT)

Abstract*:* Umpire decision making process in a very complex and important process in cricket game. When addressing the complexity of the umpire decision making process of the cricket scenarios, this research introduce a Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) based approach to classify the decision using computer vision techniques. This approach effectively combines CNNs and LSTM networks to analyze the visual data that provided the complex scenarios in cricket matches. The CNNs network used for extracting the special features from the provided short video frames such as player movement, ball trajectory and other relevant visual cues. These extracted features are then passing to the LSTM network layer that are captured the temporal dependencies and contextual information over time, enabling accurate predictions of umpire decisions. The combination of the CNNs and LSTM networks provides a powerful approach for automating decision making in cricket, improving consistency and efficiency while reducing human error and bias. This model has the potential to enhance the fairness and transparency of cricket umpiring by providing real-time, automated decision support. The integration of the CNNs and LSTM allows the model to not only recognize individual events but also understand the context over time, making more reliable and accurate decisions. This approach provides a robust solution for classifying decisions such as Leg Before Wicket, no ball and wide ball identification. By automating the umpire decision making system in cricket aims to reduce human errors, bias and improve the consistency in umpire, offer real-time decision support, ultimately enhancing the fairness and transparency. In this study, with 640 short videos training dataset and 160 short videos testing dataset the model achieved over 95% with 0.02 loss. This result demonstrated strong performance and generalization, effectively learning both spatial and temporal features of the data.

Paper ID: 1368

Empowering Accessibility Through Eye-Gaze-Driven Interaction and Real-Time Blink Detection

Sarthak Phulari (Pimpri Chinchwad College of Engineering)*; Priya Surana (Pimpri Chinchwad college of Engineering); Bhushan Pisal (Pimpri Chinchwad College of Engineering)

***Abstract**: —*In a world where accessibility and inclusionare paramount, assistive technologies are pivotal in addressing thechallenges faced by individuals with mobility and communicationdisabilities. We propose the implementation of an innovativesolution that leverages eye gaze tracking and advanced imageprocessing to enable seamless interaction with devices and environments. This technology empowers users to perform taskssuch as generating text, navigating interfaces, and controllingphysical systems, fostering independence and enhancing qualityof life. Having developed a working prototype, we are poisedto scale this transformative solution, demonstrating its precision,adaptability, and user-centered design. By bridging technologicaladvancements with human needs, this work highlights the vastpotential of assistive solutions to create a more inclusive andempowered future for individuals with disabilities.

Paper ID: 1395

Intelligent Resume Parser and Job Matching System

Rutika Bari (Vishwakarma Institute of Technology)*; Atharv Bapat (Vishwakarma Institute of Technology); Keyur Barde (Vishwakarma Institute of Technology); Shantanu Bhalke (Vishwakarma Institute of Technology); Pushkar Joglekar (Vishwakarma Institute of Technology)

***Abstract**:* The current job search ecosystem often fails to effectively align candidates with suitable job opportunities, primarily due to its reliance on job titles rather than comprehensive analysis of resume content. This limitation leads to the frequent overlooking of critical qualifications and skills, resulting in missed opportunities for both job seekers and employers. To address this issue, we propose a system that extracts key skills and attributes directly from resumes and matches them with relevant job profiles. By integrating advanced keyword extraction techniques and leveraging web scraping on platforms such as Naukri and LinkedIn, the system provides personalized and targeted job recommendations. This approach aims to enhance the efficiency and accuracy of the job search process, fostering a more meaningful connection between candidates and potential employers.

Paper ID: 1419

Reality vs. Simulation: Evaluating Machine Learning Model Performance for Predictive Maintenance on Certified and Synthetic Datasets

Priya B K (Amrita school of engineering)*; T K Ramesh (Amrita School of Engineering); Aravind R (Amrita School of Engineering); Riya Jaiprakash Khuba (Amrita School of Engineering); S Venkata Tharini Abhinaya (Amrita School of Engineering)

***Abstract**:* Predictive maintenance is an important aspect of eliminating the failure of industrial equipment, ensuring opera- tional efficiency, and reducing downtime, and machine learning in this process. In this study, a comparative analysis of machine learning model performances was carried out, based on training and evaluation, for a synthetic data set and valid certified data set obtained from NASA. Key performance metrics, such as coef- ficient of Determination i.e. R-Squared, Root Mean Square Error and Computational Efficiency, were analyzed across various regression models, including Support Vector Machines, Random Forest, k-Nearest Neighbors and long short term memory. Results revealed substantial variations in terms of model performance and consistency, highlighting how much the authenticity of the dataset influences machine models outcomes. The major concern of the study is how the dataset quality ensures trustworthy predictions from machine learning by making the importance of an origin from a synthetic dataset known with possible drawbacks in real-life applicability of its dependence.

Paper ID: 1443

DeepStressNet: Deep Learning based Multichannel EEG-based Stress Detection System

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***Abstract**:* Stress detection is vital for analyzing the mental and physical health og the person for improving the lifestyle. The deep learning based stress detection using EEG has significantly boosted automatic stress detection. However, the stress detection is challenging because of the lower long-term correlation, inferior temporal connectivity, poor accuracy for noisy signals, and lower distinctiveness of EEG features. This paper presented the stress detection using a novel DeepStressNet, which combines the deep convolutional neural network (DCNN) and long short-term memory (LSTM). The DCNN offers the spatial connectivity in multichannel EEG data, and the LSTM offers the long-term temporal depiction in the EEG signal. The proposed DeepStressNet offers improved accuracy over traditional stress detection methods. The propped DCNN-LSTM provides the recall of 91.3%, precision of 92.65, F1-score of 91.8%, accuracy of 92%, selectivity of 92.7% and negative predictive rate (NOR) of 92.6%.

Paper ID: 1472

Smart Gloves For Sign Language Translation To Text And Speech

Nida Samreen (PES University)*; M Mohana (PES University); Maneesh K M (PES University); N Manojkumar (PES University); Dinesh Singh (PES University)

***Abstract**:* For those with speech and hearing impairments, communication limitations pose serious difficulties in day-to-day interactions. By creating a wearable assistive technology that translates sign language gestures to text and speech in real time, the Smart Gloves project seeks to solve this problem. An Inertial Measurement Unit (IMU) and flex sensors are integrated into the system to capture hand orientation and finger movements. This sensor data is gathered by an ESP8266 microcontroller and sent via Wi-Fi to a server, where machine learning algorithms interpret it by comparing it with an existing dataset. Once the gesture is identified, the corresponding value is sent to the mobile phone for text and audio delivery. A linked system comprising the ESP8266, server, and mobile phone enables precise and effective gesture recognition. The suggested system is a convenient way to close the communication gap between sign language users and non-sign language users since it is scalable, portable, affordable, and user-friendly.

Paper ID: 1627

Micro-Inverter Based Small IPS on Single Load

Abdullah Mohammed Abdul Matin (Independent University, Bangladesh); Arif Ul Hoque (Independent University, Bangladesh); Utpal Chandra Mondol (Independent University, Bangladesh); Monjurul Islam Pranto (Independent University, Bangladesh); Mahady Hasan (Independent University, Bangladesh); Mohammad Rejwan Uddin (Independent University, Bangladesh)*

***Abstract**:* As the demand for reliable power sources rises, the need for efficient backup solutions becomes increasingly crucial. This paper introduces a Micro-inverter Based Small Inverter Power System (IPS) built to produce seamless power backup. The system uses micro-inverters, a compact modular technology that converts direct current (DC) from batteries to alternating current (AC) suitable for powering essential loads. There is also the use of microprocessors in this system that offer intelligent control and monitoring capabilities. The Arduino microprocessor facilitates precise regulation of inverters.

Paper ID: 1631

Solar Powered Oil Extractor with Rotating Emery Roller

Ahsanul Amin Shanto (Independent University, Bangladesh); Shahriar Amin Ronok (Independent University, Bangladesh); Minhaj Ibna Amin (Independent University, Bangladesh); Mahady Hasan (Independent University, Bangladesh); Mohammad Rejwan Uddin (Independent University, Bangladesh)*

***Abstract**:* This paper presents an innovative approach to an IoT and solar-based oil extractor system. The objective of this project is to develop an environmentally friendly and energy-efficient solution for oil extraction processes. The paper explores the design, implementation, and analysis of the system. The paper begins by discussing the motivation behind this project, highlighting the need for sustainable and renewable energy sources in the oil extraction industry. It emphasizes the importance of IoT technology in optimizing the extraction process and reducing environmental impact. The methodology employed involves the integration of solar panels, sensors, and a control system to monitor and regulate the extraction process. simulator is developed to simulate and analyze the system's performance under various operating conditions, providing valuable insights for system optimization. The analysis method encompasses evaluating energy efficiency, oil extraction yield, and overall system performance. Key findings from the analysis reveal the system's ability to operate autonomously using solar power, reducing reliance on fossil fuels and minimizing carbon emissions. The paper also highlights the importance of real-time monitoring and control enabled by the IoT aspect, resulting in improved process efficiency and reduced downtime. Overall, this paper demonstrates the feasibility and benefits of an IoT and solar-based oil extractor system, presenting valuable insights and findings for researchers, engineers, and industry professionals interested in sustainable oil extraction technologies.

Paper ID: 1632

Sortify- A 6 DOF Autonomous Voice Controlled Robotic Arm for Object Detection and Sorting using ROS2

Suhani Chalmeti (Pimpri Chinchwad College of Engineering and Research)*; Atharva Kale (Pimpri Chinchwad College of Engineering and Research); Rupali Kawade (Pimpri Chinchwad College of Engineering and Research)

***Abstract*:** This paper explores the development and integration of SORTIFY, an autonomous, voice-controlled 6-degrees-of-freedom (DOF) robotic arm designed for object detection and sorting. The research highlights the combination of key technologies including computer vision, voice recognition, inverse kinematics, and ROS2 middleware to enable precise and interactive robotic control. Using Google Speech-to-Text for voice command processing and Natural Language Processing (NLP) for keyword extraction, the system allows for intuitive, hands-free control. The robotic arm utilizes a YDLiDAR HP60C RGB-D camera for real-time color and shape detection, and depth estimation is employed to calculate the object's 3D coordinates. The system's motion control is achieved through inverse kinematics, calculating joint angles required for picking and placing objects. The integration of these technologies within the ROS2 framework facilitates seamless communication across subsystems, allowing efficient task execution. This work contributes to the field of autonomous robotics, showcasing the potential for scalable, modular, and user-interactive robotic systems in automation and industrial applications. Future research directions are proposed in areas such as machine learning for improved perception and real-time processing for dynamic environments.

Paper ID: 1647

TraumaCare: Leveraging Mobile Technology for PTSD Care and Support

Darshi Gardiya Hewage (SLIIT)*; Imesh Vitharana (SLIIT); Dinithi Mendis (SLIIT); Kimuthu Gamage (SLIIT); Sanjeevi Chandrasiri (SLIIT); Tharushi Rubasinghe (SLIIT)

***Abstract*:** Post-Traumatic Stress Disorder (PTSD) is a debilitating mental health condition that necessitates continuous monitoring for effective management. However, traditional PTSD care faces limitations such as inadequate real-time symptom tracking, limited accessibility, and a lack of integrated patient feedback. To address these challenges, this study explores a PTSD management system integrating machine learning, IoT, and natural language processing (NLP) techniques. A one-dimensional Convolutional Neural Network (CNN) enables real-time speech-based emotion recognition, allowing patients to self-assess distress levels. Additionally, NLP-based sentiment analysis of journal entries, supported by a summarization model and Optical Character Recognition (OCR), provides insights into emotional states. A two-dimensional CNN is employed for facial expression analysis, detecting stress levels and offering adaptive coping strategies. Furthermore, IoT sensors and an Arduino-based system, coupled with a Random Forest algorithm, monitor sleep patterns, heart rate, and environmental conditions, generating comprehensive reports for medical professionals. Experimental results demonstrate that integrating these technologies significantly improves symptom tracking, enhances patient engagement, and facilitates effective doctor-patient collaboration. The findings highlight the potential of IoT-driven solutions in making PTSD management more accessible, adaptive, and responsive, ultimately leading to better long-term patient outcomes.

Paper ID: 1650

Biomechanical Analyse For Weightlifting Performance Enhacement

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***Abstract*:** Weightlifting is a biomechanically demanding sport requiring precision, balance, and optimal coordination. Despite rigorous coaching, athletes often face technique flaws, injury risks, and suboptimal recovery due to human limitations in observation and feedback. This paper presents an integrated AI-powered system combining video analysis, 3D modeling, pose estimation, injury prediction, fatigue monitoring, personalized meal planning, and rehabilitation tracking. The approach leverages deep learning, machine vision, and mobile applications to deliver real-time, actionable feedback for athletes and coaches. By addressing technical performance, physical conditioning, and recovery needs simultaneously, this system aims to improve performance longevity, reduce injury rates, and introduce an innovative technological shift in sports science. The findings demonstrate high accuracy in posture detection, movement classification, and personalized recommendations, supporting the potential of AI-driven systems in revolutionizing elite weightlifting training.

Paper ID: 1663

Automated Map Generalization and LULC Classification using Machine Learning Techniques

Amritha Pillai (Amrita Vishwa Vidyapeetham)*

Abstract: Generalization of maps is a crucial task in cartography, par-ticularly for the simplification of intricate geospatial data in urban planning scenarios. In this paper, we introduce a novel hybrid framework that combines GIS-based preprocessing with machine learning-based generalization methods to process and analyze the Thampanoor area, a highly populated region in Trivandrum, Kerala. The mapping part is done using GIS tools, and the resulting spatial data was used for generalization.Roads, buildings, and boundaries were digitized and divided into 16 zones for focused analysis. In contrast to current techniques that use one technique, our method integrates the Visvalingam–Whyatt algorithm for simplification of lines, K-Means clustering for aggregation, and Chambolle's algorithm for smoothing. We also propose a typification strategy based on geometric rules augmented by Gaussian matrix classification, assessedby Silhouette Scores. Following generalization, Land Use Land Cover (LULC) classification was carried out to recognize water bodies, urban built-up, vegetation, and barren land, and class accuracy was estimated based on a confusion matrix. Our method demonstrates it successfully compromises visual simplicity and spatial accuracy and presents a scalable and sustainable method for urban planning.

Paper ID: 1675

A Data-Driven Approach to Predict Supply and Demand of Tea Types in Sri Lanka

Hashini Herath (Sri Lanka Institute of Information Technology)*; Sachin Dinujaya (Sri Lanka Institute of Information Technology); Movindu Ramsith (Sri Lanka Institute of Information Technology); Kulan Sanjula (Sri Lanka Institute of Information Technology); Lokesha Weerasinghe (Sri Lanka Institute of Information Technology); Thamali Kelegama (Sri Lanka Institute of Information Technology)

Abstract: Sri Lanka's tea industry plays a crucial role in its economy, yet the supply and demand dynamics remain unpredictable due to reliance on traditional forecasting methods. This research presents a data-driven approach integrating Google Trends, social media analytics, economic indicators, and sales data to enhance tea demand and supply predictions. By leveraging statistical and data-driven techniques, we developed a predictive framework that accurately forecasts demand fluctuations, export trends, and production levels. Our findings indicate that consumer interest, climate conditions, and economic indicators significantly influence tea trade patterns. The proposed system, implemented as a web-based dashboard, provides real-time insights for stakeholders, optimizing inventory management, production scheduling, and market strategies. This study contributes to the digital transformation of Sri Lanka's tea industry, enabling data-driven decision-making for producers, exporters, and policymakers.

Paper ID: 1684

Design of a Stacked Compact Modified Octagonal On Chip Inductor Using SKILL and EM Solver for the Advanced 5G Applications

Naveen Manchala (National Institute Of Technology Warangal)*; Bheema Rao Nistala (National Institute Of Technology Warangal)

Abstract: This paper proposes a stacked compact modified octagonal on-chip inductor with high inductance and low on-chip area. The design and simulation of the proposed inductor are simulated using an EM solver in Cadence Virtuoso. The stacked compact modified octagonal inductor shows an improvement of 3.87 times over the compact modified octagonal inductor in terms of inductance. Compared to the existing inductors in GPDK045 nm technology, the proposed stacked compact modified octagonal inductor reduced the on-chip area by 82.63% and has a quality factor (Q) above 10 over the frequencies (25-50) GHz frequencies, making it suitable for mid-5G applications.

Paper ID: 1706

Machine Learning-Based Disease Prediction and Lifestyle Management System

Shakya Jayathissa (Sri Lanka Institute of Information Technology (SLIIT))*; Pawara Hasamal (Sri Lanka Institute of Information Technology (SLIIT)); Samitha Dhnanjaya (Sri Lanka Institute of Information Technology (SLIIT)); Uresha Hansani (Sri Lanka Institute of Information Technology (SLIIT))

Abstract: Rapidly developing world, people with their daily busyness have less interest and attention towards health. Therefore, many people are at risk of developing chronic diseases and are not motivated to diagnose and treat them at an early stage. As a remedy to that problem, this paper reports an intelligent solution to help identify personalized diseases at an early stage and maintain their daily health level. The aim of this research is to bring results using their blood reports targeting people between the ages of 18-40. Accordingly, the basic purpose of this is to read the data reports to predict the diseases risk level of the individuals, recommend suitable diet plans for each individual, and recommend relevant exercises according to the BMI level and risk diseases. Similarly, the final results of this paper are to create a mobile application integrating the latest technology of the modern world with personal lifestyle to check the level of mental stress using the smart watch and accordingly predict the suggested mental wellness exercises. It should be said that this is a creative and responsible role in this prediction, which is done by mining various data. Finding the precise match between in the data set and the end prediction outcomes is the goal of the suggested system, which will be created utilizing a variety of machine learning methods. The proposed system was tested with a sample group and showed high accuracy in disease risk prediction, and the system received positive user feedback.Key Words—Chronic diseases, LMS, Optical character recognition, Image recognition, Risk prediction

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Paper ID: 1726

Analysis and Implementation of Discrete Hartley Transform using Recursive Algorithm in Verilog

Ananya Verma (Delhi Technological University)*; Priyanka Jain (Delhi Technological University)

Abstract: The paper presents the implementation and analysis of a recursive Discrete Hartley Transform (DHT) algorithm for input sequences of length N, where N = 2m and m ≥ 2, using Verilog HDL. Knowing that there has been a growing demand for high speed and resource efficient systems there is a need for implementations that are both fast and lightweight, so the proposed work aims to design a hardware optimized version of the DHT using the recursive algorithm with an IIR filter based structure to improve performance and reduce complexity. The simulation and synthesis results from MATLAB Simulink validate the correctness of the modules and functions applied and the performance of the design. The proposed architecture suits real-time signal processing systems such as radar communication and imaging. The referred recursive structure is divided into input buffering and recursive coefficient computation modules. The proposed methodology is applied in Verilog to check its feasibility. The results show that the structure is efficient in computing the DHT coefficient with minimum clock cycles, input-output ports, hardware complexity, and power consumption which are highly useful in parallel VLSI implementations.

Paper ID: 1727

Design of Octagonal Split Inductor using Parameterized Cell Methodology for 5G Applications

Sindhupriya Taduri (National institute of technology,warangal)*; Bheema Rao N (National institute of technology,warangal); Sarangam K (National institute of technology,warangal)

Abstract*:* Inductors play a crucial role in building blocks of radio frequency intigrated circuits (RFICs). High quality factor (Q) and inductance (L) values are essential for wireless applications such as mobile phones, wireless networks, and RF/microwave integrated circuits. In this paper, an octagonal split inductor is proposed. The proposed inductor is designed based on a multipath technique using parameterized cell (PCell) methodology in gpdk 45nm CMOS technology. Simulations are performed using the Cadence EMX solver. The multipath mechanism mitigates the conductor losses, thereby enhancing the quality factor and inductance value, resulting in a higher Figure of Merit (FoM). The proposed inductor gives better inductance of 3.28% and quality factor of 7.69%, against the planar octagonal inductor with the same on-chip area and Self-Resonant Frequency(SRF).

Paper ID: 1731

The Impact of Information Abundance Due to Artificial Intelligence (AI) in Engineering Education

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Abstract*:* Integrating Artificial Intelligence (AI) in higher education has significantly transformed the learning process, offering new opportunities while presenting notable challenges. This study investigates the effects of AI-driven information abundance in engineering education, particularly on technical skills, cognitive abilities, and interpersonal development. A structured survey among faculty and students provides insights into emerging trends and perceptions.Findings reveal a strong reliance on AI tools that, while improving efficiency and access to personalized learning, may also impede critical thinking, problem-solving, and collaboration. The paper emphasizes the importance of a balanced approach to AI integration, advocating for structured assessments, ethical guidelines, and comprehensive strategies to support a holistic educational experience.As AI and the Internet continue to reshape higher education, the study explores the dual nature of information abundance—its capacity to enhance learning and the challenges it introduces. Though AI facilitates technical skill development and offers tailored learning experiences, concerns arise over declining cognitive engagement, weakened teamwork, and overdependence on automation.Survey responses highlight both appreciation for AI's technical benefits and concerns over its negative impact on social interaction, engagement, ethical awareness, and classroom behavior. In response, the paper proposes strategies including responsible AI use, improved assessment methods, and a renewed focus on soft skills, communication, and collaboration.Ultimately, the study advocates for preserving the human element in education to ensure that future engineers are not only technically proficient but also ethically grounded, socially aware, and equipped with critical thinking skills necessary to thrive in an increasingly AI-driven world.

Paper ID: 1739

IoT-Enabled Edge Computing Framework for Smart Parking System

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Abstract*:* Managing parking in urban areas has become increasingly challenging as vehicle density rises, leading to con-gestion, delays, and inefficient space utilization. Existing solutions relying on centralized systems suffer fromhigh latency and limited scalability, making them inadequate for real-time, dynamic environments. This paperproposes a smart parking system leveraging edge computing and vehicular networks within a novel three-tierarchitecture. The system integrates sensors and cameras to capture real-time parking data, processed at theedge via Roadside Units (RSUs) or On-Board Units (OBUs) in parked vehicles. By distributing computa-tional tasks based on resource availability, energy levels, and processing power, the system optimizes resourceutilization, reduces RSU load, and enhances continuity. Real-time updates are delivered via a mobile/webapplication, while historical data is stored for strategic analysis. This approach improves scalability, respon-siveness, and efficiency, offering a practical solution for smart cities.

Paper ID: 1771

Automated Medicine Classification Using Deep Convolutional Neural Networks

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Abstract*:* The rapid evolution of deep learning in pharmaceutical automation necessitates precise and reliable medicine recognition systems. This research leverages the VGG19 deep convolutional neural network (CNN) to automate medicine classification based on packaging images, addressing challenges in manual verification, prescription errors, and counterfeit detection. Advanced image pre-processing techniques enhance model generalization and robustness, including normalization, augmentation, and affine transformations. VGG19 effectively distinguishes visually similar medicines by utilizing hierarchical feature extraction, ensuring high classification accuracy. Performance optimization through dropout regularization and cross-entropy loss minimizes overfitting and accelerates convergence. Model efficacy is validated using accuracy metrics, confusion matrices, and ROC curve analysis, confirming its real-world applicability. The findings underscore the transformative potential of deep learning in healthcare, offering a scalable and automated solution for pharmaceutical classification while contributing to enhanced drug safety, regulatory compliance, and patient care.

Paper ID: 1834

A Blockchain-Powered Solution for Dynamic Peer-to-Peer Energy Trading

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***Abstract**:* A decentralization-based peer-to-peer (P2P) energy trading system is suggested to address the inefficiency and deficits of traditional centralized energy markets. The suggested system integrates blockchain technology and Internet of Things (IoT) smart meters to enable secure, open, and self-sufficient energy transactions between consumers (producers and users) and users. The platform eliminates the need for middlemen using smart contracts that compute dynamic energy prices automatically based on real-time supply and demand levels. The approach promotes fair pricing, optimal market behavior, and utilization of the maximum energy resource within a localized microgrid.The system architecture enables direct trading of energy, secured by blockchain's immutability and traceability features, with IoT devices providing continuous data on energy generation and consumption. The platform is deployed using an end-to-end devel- opment stack: Ethereum serves as the blockchain base; Hardhat, Ganache,Truffle, and Remix are utilized for contract development and local emulations; MetaMask and Infura manage wallet interaction and blockchain communication; and Oracle,Solidity, Web3.js, HTML, CSS, and JavaScript enable frontend integra- tion and real-time user interaction. This integrated framework not only supports energy autonomy and decentralization but also makes the process of energy trading transparent,scalable, and secure.By combining sophisticated blockchain technology with real-time IoT data, the system has the potential to disrupt energy markets with decentralized infrastructure . Index Terms—Block chain, P2P energy trading, Dynamic Pricing, Ethereum, Smart contract, Energy Market

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Paper ID: 1839

Integrating Digital Twin Models for Personalized Treatment Strategies in Modern Healthcare

Jayabhaduri Radhakrishnan (Alliance University)*; Shekhar R (Alliance University)

***Abstract**:* Digital Twin (DT) technology has become a revolutionary innovation in contemporary healthcare, allowing real-time, patient-specific modeling and simulation for enhanced clinical decision-making. The current study reports the design and deployment of a Digital Twin framework for customized treatment planning with emphasis on oncology treatment. Integrating disparate patient information—clinical records, diagnostic imaging, and physiological measurements—the Digital Twin replicates disease progression and predicts treatment outcomes specific to a patient's profile. The model uses sophisticated predictive analytics to benchmark conventional treatment protocols against optimized, individualized approaches formulated by the Digital Twin. Its performance was validated through real-world clinical datasets to provide a high level of predictive precision, recording an average Mean Absolute Percentage Error (MAPE) of 8.4% in forecasting tumor volume. In addition, the optimized treatment plans achieved an average 12.8% tumor reduction and 16.3% side effect severity reduction against standard methods. Simulation runtimes were on average 13.7 minutes, rendering the solution feasible for near real-time clinical assistance. These findings highlight the promise of Digital Twin technology to enable more accurate treatment, minimize side effects, and enhance patient outcomes overall. Future efforts will investigate integration with real-time sensing data as well as extension to other chronic disease areas, facilitating fully adaptive, smart healthcare systems.

Paper ID: 1840

Machine Learning Based Visual Style Advisor for Personalized Hair Color Recommendation

Devni Tharushika (SLIIT)*; Devni Tharushika (SLIIT); Manori Gamage (SLIIT); Uditha Dharmakeerthi (SLIIT)

***Abstract**:* Hair color is a major aspect of an individual's style and self-expression which can impact their appearance and self-confidence. However, identifying the right hair color can be complicated without a professional's help or knowing details about skin tone and eye color. The irreversible nature of hair color application leads to wastage of money and frustration when wrong choices are made, making accurate simulation essential. This paper proposes a "Machine Learning Based Visual Style Advisor for Personalized Hair Color Recommendation" to support user decision-making by enabling informed choices through real-time simulation. The proposed simulation system allows users to preview potential outcomes without physical alteration, eliminating costly mistakes. The method uses facial feature extraction with computer vision to utilize nearly all facial attributes in recommendations. Skin color and eye color are detected using MediaPipe Face Mesh and OpenCV, with dominant colors identified through K-Means Clustering. User data including gender, hair type, personal preferences and occasion aid in accurate recommendations. Hair segmentation using ResNet18 isolates the hair area, allowing color prediction to be applied only to hair without altering other facial features. A deep learning segmentation model (BiSeNet) and color mapping logic enhance precision. The hair color prediction model using XGBoost and Synthetic Minority Over-sampling Technique class balancing achieved 100% accuracy, demonstrating system reliability for contextual, real-time recommendations. By combining machine learning, deep learning and computer vision, the system prevents irreversible hair color mishaps through accurate simulation while reducing dependency on professional consultation.

Paper ID: 1849

CattleSite : Integrated Veterinary Application for Enhanced Cow Health Management

Poornaka Perera (Sri Lanka Institute of Information Technology); Vihigumie Hettiarachchi (Sri Lanka Institute of Information Technology)*; Thesarana Dissanayake (Sri Lanka Institute of Information Technology); Dasunika Ekanayake (Sri Lanka Institute of Information Technology); Buddhika Harshanath (Sri Lanka Institute of Information Technology); Manori Gamage (Sri Lanka Institute of Information Technology)

***Abstract**:* The research "Integrated Veterinary Application for Enhanced Cow Health Management" concentrates on exploiting cutting-edge technologies for enhancing the dairy cow farming practices confronting major obstacles experienced by veterinarians and farmers, which include complications in early disease detection, finite access to reliable diagnostic tools, inadequate tracking of health and nutritional data, absence of procedures to prognosticate the future milk yield, and lack of proper methods to reach an available veterinarian in an emergency. The proposed system comprises the Internet of Things (IoT), Machine Learning (ML), and centralized data management to elevate the cow care practices by providing real-time monitoring, personalized nutrition recommendations, and automated alerts for vaccinations and check-ups. Early disease detection features exploit anomaly detection and pattern recognition to identify possible health issues, allowing timely interventions by obviating critical situations and reducing treatment costs. Additionally, by the usage of Random Forest technique, a predictive model permits accurate milk yield forecasting based on historical data and environmental factors. Furthermore, this research presents a cow health management system that aims to enhance the accessibility and effectiveness of veterinary care. This feature provides the capability of detecting nearby available veterinary clinics with good services, facilitating communication between farmers and veterinarians for urgent consultation, appointment scheduling with an alert system, and veterinary profile management. The ultimate outcome of this innovative approach upgrades data-driven decision-making, resource allocation, and operational efficiency, facilitating the development of a profitable and sustainable dairy farming industry.

Paper ID: 1850

Theoretical Investigation and Comparison of Different Plantar Pressure Measuring Devices

Dr. Gaurav Bharti (Indian Institute of Information Technology Bhopal)*

***Abstract**:* Plantar pressure analysis is crucial for a detailed visualization of the health problems regarding the foot. A lot of devices lack in efficient analysis and effective management for testing a patient. The paper covers a detailed theoretical investigation of various Plantar pressure measuring device with the proposed model based on the principle of Fibre Bragg Grating (FBG). The system tests the patient within time of less than 80-90% of its predecessor and proposes a lot of new functions such as record of patient data, and its transfer to other devices. The paper covers the future scope of the research for the proposed device.

Paper ID: 1869

Innovative Enhancements in Online Delivery Service

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***Abstract**:* The growth in online delivery services and ecommerce has increased the demand for accurate, efficient, andreliable courier services. However, problems with unstabledelivery time, inefficient workforce management, inefficientroute planning, and unverified customer feedback compromiseoperational effectiveness and customer satisfaction. Thisresearch proposes an integrated solution based on ArtificialIntelligence (AI), Machine Learning (ML), Natural LanguageProcessing (NLP), and predictive analytics to optimize courierservice operations. The research proposes four integratedsystems: Automated Voice Activation System through NLP andTwilio for real-time delivery status updates, Data analytics driven workforce management system through ML foroptimizing employee performance and staffing requirements,Fleet management solution through real-time traffic andweather-based route optimization and Voice sentiment analysisand image-based parcel verification AI-driven customerfeedback verification system. These systems are designed tominimize inefficiencies, optimize strategic processes, andimprove quality of service in general. The processes include AImodel creation and building, ML algorithm training usinghistorical delivery and staff data, voice and image processingintegration for complaint verification, and performancetracking with real testing. Primary metrics of delivery accuracy,response time, staff productivity, and customer satisfaction aretracked for assessing the impact. The results of the systemimplementation should show better estimation of delivery times,increased productivity of employees, fewer delays, and moreaccurate customer feedback confirmation. Through thecombination of cutting-edge technologies, this study offers anautomated and scalable solution to logistics issues, creating anew benchmark for efficiency in the courier industry.

Paper ID: 1870

Cyber Security Auditing Using Ai Tools

Prajwal Landage (B.M.S College of Engineering.)*; Prajwal Landage (BMS College of Engineering)

***Abstract**:* Organizational defense measures now must includeCybersecurity auditing, particularly in light of emerging threatslike

Denial of Service (DoS) and Man-in-the-Middle (MitM)attacks. Conventional auditing methods frequently don't respondin real-time or adjust to changing danger scenarios. In orderto work with real-time data for automated threat detectionand compliance validation, this article suggests an AI-drivencybersecurity auditing architecture that is in line with ISO/IEC27001 standards. For effective network traffic classificationand anomaly detection, the system uses supervised learningtechniques, particularly Support Vector Machines (SVM) andDecision Tree (DT) algorithms. In order to guarantee ongoingsecurity compliance, the framework also includes ISO/IEC 27001 policy checks. This paper presents the suggested architecture, methods, and pertinent research background, even though the implementation is still proceeding. Live deployment, assessment, and extension to multi-domain threat detection with additional AI models are all part of the future development.

Paper ID: 1880

Development of an Integrated Machine Learning System for Personalized Thyroid Disease Management: Combining Diagnostic Prediction, Risk Assessment, and Dietary Optimization

Samali Hapugahapitiya (Sri Lanka Institute of Information Technology)*

***Abstract**:* If not diagnosed and treated, thyroid disease, such as hyperthyroidism, hypothyroidism, and structural abnormalities, leads to serious health complications. Individualized treatment plans are required to obtain the best patient results and avoid the complications of inappropriate or delayed treatment. For better risk stratification, treatment planning on an individual basis, and diagnostic accuracy, the study seeks to create a machine learning-based personalized treatment recommendation system for thyroid disease. Thyroxine antibody tests (TPOAb, TgAb, TSHRAb), demographics, clinical history, medical imaging, and diet are a few of the many sources of data that are integrated in this study. Neural networks, CNN Inception V3, Random Forest, and other machine learning algorithms are used for prediction and classification. Based on thyroid status and comorbidities, a natural language processing (NLP) module and a multilayer perceptron classifier create personalized diet plans. In thyroid dysfunction detection, high-risk patient identification, and personalized therapy recommendations, the suggested approach demonstrated excellent accuracy. The proposed method enabled early diagnosis and improved diagnostic accuracy substantially using machine learning. The results demonstrate the promise of data-driven approaches to personalized treatment of thyroid disease. This platform enhances patient outcomes and adherence to medical regimens by harnessing the power of machine learning and clinical data to assist in early diagnosis, optimization of treatment plans, and individualized nutrition planning.Keywords— Thyroid disease, Machine learning, Risk assessment, Dietary recommendations.

Paper ID: 1885

Integrated Mobile Speech Therapy Platform for Child Language Development and Progress Monitoring

ishuwara Lianaarachchi (Sliit)*; Hashan Madusanka (Sri Lanka Institute of Information Technology); Chathumin Herath (Sri Lanka Institute of Information Technology); Hashani Rathnasekara (Sri Lanka Institute of Information Technology); Lokesha Weerasinghe (Sri Lanka Institute of Information Technology); Akshi De Silva (Sri Lanka Institute of Information Technology)

***Abstract**:* Speech therapy is a critical field dedicated to diagnosing and treating speech and communication disorders, primarily managed by Speech-Language Pathologists (SLPs). It plays a crucial role in addressing articulation disorders, language delays, and impairments caused by conditions such as cleft palate and hearing loss. However, in Sri Lanka, the effectiveness of speech therapy is often compromised by insufficient parental engagement due to factors such as time constraints, low literacy levels, and limited technical knowledge. These challenges hinder therapy accessibility and consistency, particularly in rural and undeserved areas. This study presents an Integrated Mobile Speech Therapy Platform, designed to enhance the efficiency and accessibility of speech therapy for children through AI-driven innovations. The platform incorporates real-time feedback, progress monitoring, and voice recognition technology to improve pronunciation and articulation exercises. By transforming traditional speech therapy methods into interactive and engaging activities, the system encourages active participation, fostering continuous practice and personalised treatment plans tailored to each child's specific needs. A core component of this platform focuses on interactive and child-friendly therapy activities, specifically designed for children aged one to three, a critical period for language acquisition. Traditional methods often struggle to maintain engagement, leading to sub-optimal results. The proposed solution integrates pronunciation exercises, vocabulary building games, and fluency-enhancing activities, featuring an echo function, real-time audio-visual feedback, and a recording and playback tool to reinforce learning and self-evaluation. Built on an adaptive learning framework, the system personalises therapy based on initial assessments and continuous evaluations, dynamically adjusting the difficulty of exercises to match the child's progress.

Paper ID: 1898

A Survey on Secure Predictive Healthcare Systems: Integrating Blockchain, IoT Sensors, and Deep Learning Models

Priyanka Dhumal (Dr. D. Y. Patil Institute of Technology)*

***Abstract**:* The role of healthcare technology within the early detection and prediction of symptoms of diseases has evolved due to the amalgamation of Blockchain, Internet of Things (IoT), and Deep Learning (DL) technologies. This survey aims to highlight the integration of these technologies toward devising an efficient, secure, intelligent, real-time monitoring system for health care applications. Remote monitoring of Overall physical health is now possible through continuous collection of ECG, heart rate, blood pressure, and oxygen saturation data through IoT-enabled wearable sensors. These health data streams are analysed by deep learning models—Convolutional Neural Networks (CNNs), LSTM networks, or hybrid models—deployed on edge or cloud computing platforms to identify and predict with high accuracy the early detection of systemic health issues. Blockchain technology addresses security, privacy, and integrity issues by providing tamper-proof decentralized storage, transparent access control, and controlled medical data sharing which ensures privacy. Automated smart contracts aid in the facilitation of secure data transmission and automate access permissions between patients, doctors, and healthcare institutions. This survey divides critical findings along thematical, algorithmic, and evaluation standards including accuracy, sensitivity, and latency. The results support that blockchain technology preserves data immutability and accountability, while deep learning sharpens the accuracy of diagnoses. Also discussed are the challenges of energy efficiency in IoT devices, scaling the blockchain, data heterogeneity and model interpretability, as well as proposed future avenues of inquiry such as federated learning, quantum blockchain, and cross-platform interoperability. In conclusion, this research emphasizes the integrated BIoT-DL framework that enables predictive, personalized, and preventive overall physical health.

Paper ID: 1936

Video Summarization of CCTV Footage

Pradeep Surasura (KLE Institute of Technology Hubballi)*; Shilpa E (KLE Institute of Technology Hubballi); Rajeshwari V Patil (K L E Institute of Technology Hubballi); Vishwanath Kamatar (KLE Institute of Technology Hubballi); Vanitha GP (KLE Institute of Technology Hubballi); Mahantesh Sajjan (K.L.E. Institute of Technology Hubballi)

***Abstract**:* CCTV cameras record everything in their view, often creating long videos. Most of these videos don't have the important details. Large video files take up a lot of space and need people to watch them. The latest study was found that only about 3 hours out of a 24-hour video contains meaningful motion. Video summarization helps by making a short version of the footage. It picks out key moments and removes repeated or unimportant parts. This short summary lets security staff see important events quickly without watching hours of footage. It saves time and effort, making monitoring easier. This procedure will help in areas like crime detection and healthcare.

Paper ID: 1954

Artificial Intelligence-Based Simulation and Detection of Cyber Threats in Smart Home IoT Environments- A Comprehensive Review

Rachana G (B.M.S. College of Engineering)*; Panimozhi K (B.M.S. College of Engineering)

***Abstract**:* The proliferation of smart home IoT devices has extended the attack surface and created fresh vulnerabilities, leaving such homes open to ever more advanced cyber threats. This review examines how artificial intelligence (AI) methods, including machine learning (ML) and deep learning (DL) models, are being utilized to simulate and recognize threats of this nature in real time. It explores the recent techniques in anomaly detection, intrusion prevention, and behavior analysis, and emphasizes the strengths and weaknesses of the current frameworks. It also reviews commonly adopted datasets, simulators, and threat models in the literature. Finally, open issues including scalability, data privacy, and adaptability, and future research trends on AI assisted security in smart IoT networks are discussed.

Paper ID: 1958

LLM-Driven Resume Matching with Preference Re-Ranking and Skill Recommendations

Harshitha Sheshagiri (PES University)*; Viren Rajan Mundre (PES University); Chandan L S (PES University); Suresh Jamadagni (PES University)

***Abstract**:* To address the shortcomings in existing approachesof job recommendation and skill recommendation systems, thispaper proposes a novel LLM-based job and skill recommendationsystem that performs combination of keyword matching, seman-tic analysis, and personalized preference modeling via Large Lan-guage Models (LLMs). Unlike previous approaches ,we proposea hybrid approach which combines both key word matching andsemantic matching along with user preferences such as roles,company culture, place of work, mode of work(remote ,onsite,hybrid) etc, these features are further analyzed by LLM's forpersonalized job and skill recommendations. Due to the combi-nation of keyword matching, semantic matching,fuzzy matchingand prompt-based instructions,this approach demonstrates thepotential to deliver an explainable, ideal, and personalizedjob and skill recommendation system. This approach bridgesthe gap between static resume matching and career guidance,significantly personalizing the guidance process for job seekers.Index Terms—LLM,semantic matching,fuzzy matching

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Paper ID: 1961

Intelligent Interconnect for Cache-Coherent Multicore Systems

Gurunadha Mangalampenta (Matx)*

Abstract: The increasing core count in modern multicore processors has pushed traditional interconnects to their limits, particularly in maintaining cache coherence while minimizing latency, power, and complexity. This paper presents an in-depth analysis of intelligent interconnects that enhance the performance and scalability of cache-coherent systems. We explore current limitations of conventional interconnect architectures, discuss novel intelligent techniques such as adaptive routing, machine learning-guided arbitration, and topology-aware coherence protocols, and provide comparative insights using simulations and modeling. The proposed approach demonstrates significant improvements in latency, coherence traffic reduction, and energy efficiency.

Paper ID: 1981

YP2D: A Comprehensive Resource for AI-Driven Yoga Pose Classification and Alignment Solutions

Ashutosh Kumar (Thapar Institute of Engineering and Technology)*; Vinay Kumar (Thapar Institute of Engineering and Technology); Madhur Verma (Thapar Institute of Engineering and Technology); Mihir Pradhan (Thapar Institute of Engineering and Technology); Pratham bathla (Thapar Institute of Engineering and Technology); Mukesh Dalal (Thapar Institute of Engineering and Technology); Ashish Gupta (Thapar Institute of Engineering and Technology)

Abstract: Yoga, an ancient practice from India, mentioned in Rig Veda, is famous for promoting physical, mental, and spiritual well-being. However, achieving correct posture alignment is a challenging task, especially in self-guided practice, as incorrect alignment reduces its effects and may lead to body harm like joint stress and muscle strain. To address these issues, this paper introduces the proposed Yoga Pose Perfect dataset (YP2D), a well-structured resource to support the development of AI-driven yoga posture detection and correction systems. The dataset consists of 3,873 annotated images of 10 Yoga Asanas captured from three distinct angles—front, right, and left. It includes annotations for 33 body landmarks, bounding boxes, and corrective feedback, covering practitioners of different body types. YP2D is tested using MediaPipe Module and offers various real-world practical applications. YP2D establishes a strong foundation for advancements in fitness platforms, healthcare, and AI-driven pose alignment tools, as it offers inclusive, well-annotated, and diverse poses. To the best of the authors' knowledge, this is one of the first and finest datasets to include discussion on inclusivity, detailed annotations, and pose variations, bridging India's ancient knowledge with modern AI Technology to promote better and safer practices globally.

Paper ID: 2018

A Resilient Optical Character Recognition Architecture Powered by Artificial Neural Networks (ANNOCR) for Automated Vehicle Plate Identification

Praveen Yadaw (Kalinga UNIVERSITY)*; Soma Rajwade (Kalinga University); Akash Ingle (Kalinga University); Ms Aakansha Soy (Kalinga University); Anu G Pillai (Kalinga University); Lekhraj Sahu (Kalinga University)

Abstract: Automatic Number Plate Recognition (ANPR) constitutes a pivotal component of intelligent transportation systems (ITS) and automated surveillance architectures. This study presents a computationally efficient framework for robust localization and recognition of Indian vehicular license plates, employing a hybridized image processing pipeline executed in MATLAB. The system operates on RGB imagery of dimension 158×319×3 and performs sequential operations comprising intensity normalization, grayscale transformation, adaptive thresholding, and morphological filtering to attenuate noise and enhance feature saliency. Sobel-based edge extraction is utilized to delineate high-gradient plate regions from cluttered backgrounds, followed by bounding box segmentation to isolate alphanumeric glyphs. A template-matching Optical Character Recognition (OCR) module facilitates accurate decoding of registration identifiers. Histogram analysis of grayscale pixel intensities (0–255) substantiates optimal contrast distribution, enhancing binarization efficacy. Empirical evaluations demonstrate the model's high precision across diverse illumination scenarios and heterogeneous plate configurations prevalent in Indian roadways. The architecture exhibits resilience to ambient perturbations and scalability for real-time deployment via embedded controllers interfaced with ZigBee wireless modules. Targeted use cases include high-security installations, governmental access points, toll collection systems, and intelligent parking infrastructure. This framework offers a low-latency, resource-conservative solution with high recognition fidelity, constituting a viable subsystem for real-time vehicular monitoring. Prospective extensions include integration with cloud-based databases for dynamic tracking, anomaly detection, and automated infraction reporting in smart urban environments.

Paper ID: 2019

Cyber Shield X: AI-Driven Cyber Defense Suite

Shreyas M (Global Academy of Technology)*; Dr. Jyothi R (Global Academy of Technology); Dr. Kumarswamy S (Global Academy of Technology)

***Abstract**:* With the rapid escalation of cyber threats in bothfrequency and complexity, conventional security tools areincreasingly inadequate. Cyber Shield X is an AI-Driven Cyberdefense suite designed to provide real-time threat detection andprevention through a modular dashboard system. The platformintegrates phishing detection, data leak monitoring, systemaudits, and threat simulation modules, all built using Python,Flask, Django, MongoDB, and Next.js. Unlike traditionalsystems, Cyber Shield X incorporates both heuristic rules andAPI-based threat intelligence, with future provisions formachine learning-based classification. This paper presents thesystem's architecture, implementation, evaluation metrics, andsimulation results, demonstrating its potential as an educationaland lightweight cybersecurity solution. Performance metrics,including accuracy and F1-score, are analyzed to validate itseffectiveness.

Paper ID: 2020

AI Powered Job Recommendation System

Vikas Shanabhog (Global Academy of Technology)*; Kanagavalli R (Global Academy of Technology); Kumaraswamy S (Global Academy of Technology); Jyothi R (Global Academy of Technology)

***Abstract**:* In today's world, finding the right job is difficult for the aspiring candidate. Present job recommendation systems face challenges when dealing with the unstructured data present in the resumes. These systems lack accuracy. The main goal of this study is to develop a system that takes structured data through forms and unstructured data that is present in resumes to provide job recommendations. This system allows users to enter their academic and professional details or upload their resume. It is trained separately for both resume and form data to give job recommendations and is integrated with Django. The user can also apply for the job by clicking on View Jobs. This dual recommendation helps to enhance the accuracy and provide seamless recommendations.

Paper ID: 2041

A Machine Learning Framework for Phishing URL Detection Using Lexical and Deep Learning Techniques

Remy Renjit (Dr. Viswanath Karad MIT World Peace University)*

***Abstract**:* Phishing represents one of the most prevalent and evolving cyber threats that commonly use maliciously crafted URLs to exploit trust in human users. Existing detection methodologies such as blacklisting find it difficult to discover newly created, obfuscated, and zero-day phishing attacks, as they are inherently reactive. As such, we present a defendable and scalable phishing detection framework based on machine learning methods. The framework integrates classical models (Random Forest and Logistic Regression) with an enhancedhybrid Deep Learning model comprising GRU and CNN, employing a majority of lexical features extracted from URLs,which include features that describe structural characteristics of the URL, identifier characteristics of the protocol, and keywords related to phishing. After consistent normalization of URLs, which enhances the output reliability for feature descriptions, our enhanced hybrid GRU-CNN model captures either sequential or ordered characteristics of the URL, resulting in enhanced recognition of URL-based phishing strategies. We evaluated the framework's efficacy by utilizing 5,000 samples, and the results demonstrated that our RF model is superior in phishing detection accuracy and is a lightweight, high-performing option for real-time incident analysis. Whereby we proposed a practical, deployable, and generalizable framework to support overall enhanced cybersecurity for browsers, mail or ad gateways, and endpoint-protection platforms.

Paper ID: 2056

Expenses Budget Management System with Chatbot Implementation USING OPEN AI

Trisha Sudhakar (Global Academy Of Technology)*; Dr. Sheela S (Global Academy Of Technology); Dr Jyothi R (Global Academy Of Technology); Dr. Kumaraswamy S (Global Academy Of Technology)

***Abstract**:* The Expenses Management System is a web application developed using the MERN stack (MongoDB, Express.js, React.js, and Node.js) to provide an efficient solution for tracking and managing personal expenses. The system allows users to streamline their financial management through features such as category creation, expense recording, budget setting, and detailed expense summaries. Users can register and log in securely to access their personalized dashboards, where they can create custom expense categories, add expenses with details like date and amount, and view their spending history. A key feature is the ability to set budget limits for different categories, enabling users to monitor their expenses against predefined thresholds. The application also generates insightful reports based on date ranges, offering a comprehensive view of spending trends. Designed with a user-friendly interface and robust back-end architecture, this system ensures seamless operation and scalability. By leveraging the capabilities of the MERN stack, the application delivers high performance, flexibility, and responsiveness, making it an ideal tool for effective financial management

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Paper ID: 2068

Ant Colony Optimization Based Feature Selection for Early Detection of Heart Disease using Combined Dataset

Shaik Shafi (B V Raju Institute Of Technology)*

***Abstract**:* In the recent past, irrespective of age limit and gender, most of the people are losing their lives due to heart disease globally. According to the survey of World Health Organization (WHO), around 17.9 million people are dying every year due to heart diseases. Prediction and tracking of unusual deaths due to sudden cardiac disease in humans is a quit challenging process that necessitates accuracy and efficiency. The chance of death will be reduced with early prediction of the disease. Towards this, there exists plethora of studies for early detection of heart disease based on Machine Learning (ML) approaches, typically relaying on single dataset However, such approaches often result in imbalanced and biased prediction. Thus, in order to address the issues, in this paper an efficient heart disease detection model that leverages various ML algorithms applied to a combination of three publicly available datasets is proposed. The proposed method makes use of ACO based feature selection algorithm to reduce the redundancy generated by large data sets and enhances the performance of the classifier in the next stage. Further, the ACO algorithm effectively identifies the best features with high convergence speed. Then, in the classification phase, various ML based algorithms-CatBoost, Naïve-Bayes (NB), Random Forest (RF), K-Nearest Neighbors (KNN) and Support Vector Machine (SVM) were employed to assess the efficacy of the selected features. Each model was trained using the feature selected dataset and subsequently tested. Among these, the SVM model combined with ACO-based feature selection achieved the highest accuracy of 98%, outperforming the other ML algorithms.

Paper ID: 2084

Denoising of Polysomnographic signals using RNN(Bi-LSTM)

Ritankar Bhattacharya (KIIT University)*; Pushpendu Kumar Mandal (KIIT University); Suramya Misra (KIIT University); Ritam Bhattacharya (KIIT University); Riya Kumari (KIIT University); Sujoy Dutta (KIIT University)

***Abstract**:* In the field of biomedical signal processing, the accu-rate analysis of Polysomnographic (PSG) recordings is essential,diagnosing sleep-related disor ders and studying neurologicalactivity. However, signals such as Electroenceph alography (EEG)and Electrooculography (EOG) collected during sleep studies areoften contaminated by noise due to muscle contractions, externalmovements, and environmental interference. These distortionshinder clinical evaluation and diagnosis. This study proposesa noise reduction method using a Bi-directional Long-Short-Term Memory (Bi-LSTM) autoencoder. Incoming signals arepre processed through bandpass filtering and interpolation topreserve key physio logical frequencies. The model is trained toreconstruct clean signals from noisy inputs, achieving low re-construction loss while maintaining essential brainwave features.Both visual comparisons and loss metrics demonstrate significantim provements in denoising performance. The proposed approachoffers a promising step towards an automatic and reliable noisereduction in PSG analysis.

Paper ID: 2124

RiceNet: An Ensemble-Based Deep Learning Framework with Grad-CAM for Explainable Rice Variant Identification

Md. Nawab Yousuf Ali (EWU)*; Golam Sorwar (Department of Computer Science and Engineering Southern Cross University Australia); Fahad Ahammed (Department of Computer Science and Engineering East West University Dhaka, Bangladesh); Omar Faruq Shikdar (East West University)

***Abstract**:* Rice is one of the major crops harvested around the world. It is a staple food in many countries of the world. There is a plethora of rice variants. Identifying the variant of rice grain is an important task for harvesting rice and creating new climate-resilient variants. The goal of this study is to leverage deep learning techniques to create a system that can detect a variety of rice on its own. A dataset of 19 thousand images comprising of 38 distinct rice variants was utilized for model training and evaluation. Three pre-trained convolutional neural networks (CNNs): DenseNet201, InceptionV3, and EfficientNetB4 were employed with EfficientNetB4 exclusively integrated into an ensemble model. The images were pre-processed prior training. To fine tune the models, some of the layers of the model were frozen and hyperparameter tuning was performed. The ensemble model achieved the highest accuracy of 99.59%, which is the best result among all the models. To enhance model interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) was integrated, providing visual explanations of classification decisions.

Paper ID: 2136

Mathematical modeling and simulation of a simplest IT2 Mamdani fuzzy PI/PD controller

Abhishek Raj (Indian Institute of Technology, Kharagpur)*; B. M. Mohan (Indian Institute of Technology, Kharagpur)

***Abstract**:* This paper presents mathematical models of a simplest Mamdani Interval Type-2 (IT2) Fuzzy ProportionalIntegral (FPI)/ Fuzzy Proportional Derivative (FPD) controller. IT2 Fuzzy Sets (FSs) with triangular Mem-bership Functions (MFs) are considered for fuzzifying scaled input variables and IT2 singletons are used foroutput variable fuzzification. Weighted Average (WA) defuzzification with Karnik-Mendel (KM) algorithm isused for finding the mathematical models of the proposed controller. Next, controller properties are examinedfor its suitability for control. To validate the applicability of the proposed controller, a simulation study isconducted on a nonlinear plant (DC series motor), and the corresponding results are compared with the resultsof linear PI and existing Type-1 (T1) FPI controllers.

Paper ID: 2140

SponsorsRisk: Longitudinal Risk Profiling of Clinical Trial Sponsors Using Multi-trial Histories

Ganesh Pathak (COEP Technological University)*

***Abstract**:* Clinical trial failures contribute to delays in drugdevelopment, financial losses, and missed opportunities for patientbenefit. Existing research largely focuses on trial-levelprediction, overlooking the sponsor's historical performance asa predictive factor. This paper presents SponsorsRisk, a sponsorcentricdeep learning framework that leverages multi-trial historiesto predict the likelihood of failure in future trials. Usingthe Aggregate Analysis of ClinicalTrials.gov (AACT) dataset,we model each sponsor as a sequence of trials ordered bystart date, capturing temporal risk patterns with GRU andTransformer architectures. Our approach integrates structuredmetadata, temporal embeddings, and sponsor-type information,enabling both accurate predictions and interpretability via SHAPand attention mechanisms. Comparative analysis across sponsorcategories reveals meaningful differences in risk dynamics. Theproposed methodology offers actionable insights for regulators,investors, and research institutions seeking to assess and improveclinical trial reliability.Index Terms—Clinical trials; sponsor risk; sequence modeling;GRU; Transformer; calibration; interpretability

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Paper ID: 2166

HealthGuardian: A Smart Platform for Health Decision Support and Personalized Recommendations

Dasini Sumanaweera (SLIIT)*; Vidumini Layangika (SLIIT); Ramod Prabashwara (SLIIT); Kavindya Rajapaksha (SLIIT); Anuradha Jayakody (SLIIT); Pipuni Wijesiri (SLIIT)

***Abstract**:* The growing prevalence of lifestyle-related health issues among adults in Sri Lanka poses a significant challenge to healthcare systems and insurance frameworks. Traditional health management solutions often lack predictive insight, personalization, and integration with insurance planning. To address these gaps, HealthGuardian introduces an intelligent, web-based platform designed to proactively monitor health compliance and deliver personalized wellness interventions. The system integrates four core components: Activity Level Assessment, Health Risk Prediction, Customized Exercise Planning, and Insurance Package Optimization. By employing machine learning techniques such as classification, clustering, and predictive modeling, the platform analyzes individual fitness data to forecast health risks, enhance user engagement, and tailor insurance offerings. Key features include real-time analytics, dynamic risk visualization, and progress tracking to drive informed decision-making and improve health outcomes. This paper details the system's architectural design, outlines functional and non-functional requirements, and proposes a roadmap for real-world validation ultimately supporting proactive health management and insurance innovation in the Sri Lankan context.

Paper ID: 2181

Enhancing Campus Navigation with Augmented Reality: A Mobile-based Directory System for a University Setting

Francis Carabuena (Technological Institue of the Philippines-Manila)*

***Abstract**:* Many large universities rely on printed maps and verbal directions. However, these often require remembering several steps and may lack clarity, leading to confusion and inefficiencies in campus navigation. This study introduces a mobile-based augmented reality (AR) navigation system designed to enhance wayfinding through QR-based location anchors, Dijkstra's algorithm for optimal path computation, and real-time 3D routing using Unity's NavMesh Agents. The system features a multi-role access model customized to navigation options for visitors, new students, and returning users to improve usability and security. A voice navigation feature is incorporated into the system to promote accessibility and inclusive use. The Rapid Application Development (RAD) methodology was used to develop an application designed to run on low-end Android devices and operate offline using local data storage. The user acceptance testing achieved a grand mean of 6.71, indicating that users strongly agree that the system offers a scalable, role-sensitive AR solution for modern educational institutions.

Paper ID: 2182

Diagnosis of Poultry Diseases from Fecal Images Using Convolutional Neural Networks

Prasannavenkatesan Theerthagiri (GITAM University Bengaluru)*; Gopala Krichnan C (GITAM University Bengaluru)

***Abstract**:* Poultry farming is among the major aspects of the world food industry, but disease outbreaks constitute huge impact on productivity and animal welfare. Early detection remains paramount in ensuring that infections do not get disseminated and retaining economic losses to a minimum. The project aims to develop a machine learning intelligent poultry disease detection system using image processing and deep learning. The system accepts poultry images and through CNN and state-of-the-art AI techniques correspondingly classify the diseases. The automated and efficient solution hence assists farmers in disease detection in the early stages, so that intervention can take place in time for better management of diseases. This leads to the welfare of the poultry, hence contributing to the sustainability of farming by reducing losses and improving output.

Paper ID: 2191

Intrusion Detection as a Service for Multi-Tenant Cloud Environments (IDaaS)

Smitha GV (Dayananda Sagar College of Engineering)*

***Abstract**:* Cloud has taken the center-stage of IT infrastructure but this growth comes with the fact that multi-tenant environments are exceptionally prone to hacking including DoS/DDoS, malware and infiltration. Traditional Intrusion Detection Systems (IDS) tend to become outdated quickly and can be resource heavy, unable to handle the specific security needs of different cloud users. This paper proposes IDaaS or Intrusion Detection as a Service, which is a cloud security model that is tenant-conscious. The proposed architecture provides heavy lightweight security as a result of virtualization-based surveillance and dynamic rule dissemination with advanced light-weighted protection. Detection is enhanced when tenants are allowed to deploy only the necessary rule sets tailored to them, thus reducing overhead. The prototype evaluation demonstrates the model as flexible, accurate, cost-efficient, and effective intrusion detection, showcasing significant potential to bolster security in multi-tenant cloud environments.

Paper ID: 2197

Metamaterial-Embedded Dual-Band Antenna with Meander Line Feeding and Truncated Ground

Suganthi J (PES University)*; Kavitha T (New Horizon College of Engineering)

***Abstract**:* Wireless communication is one of the most evolved areas of research in today's world. Any communication model consists of a transmitter end and receiver end. In this project a compact, Metamaterial based Microstrip Patch antenna operating in X band has been designed and simulated using FR4 as substrate with height of 1mm. The antenna resonates at resonant frequency of 7.4 GHz and 9.8GHz, a return loss value of -15.4 dB, -34.49dB and a corresponding gain of 3.13dB, 3.118dB. Metamaterials have been introduced on the patch and on the substrate to enhance antenna performance. Meandering feeding techniques is used in order to improvise bandwidth of the antenna more than 900MHz. Designed antenna has a good value of return loss and bandwidth has been enhanced. In order to reduce back radiation, front-to-back ratio has been changed by changing the ground plane length bring it to and optimum length. The designed antenna is simulated, and parameters have been analysed.

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